



OPEN ST-CFI: Swin Transformer with convolutional feature interactions for identifying plant diseases

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The increasing global population, coupled with the diminishing availability of arable land, has rendered the challenge of ensuring food security more pronounced. The prompt and precise identification of plant diseases is essential for reducing crop losses and improving agricultural yield. This paper introduces the Swin Transformer with Convolutional Feature Interactions (ST-CFI), a state-of-the-art deep learning framework designed for detecting plant diseases through the analysis of leaf images. The ST-CFI model effectively integrates the strengths of the Convolutional Neural Networks (CNNs) and Swin Transformers, enabling the extraction of both local and global features from plant images. This is achieved through the implementation of an inception architecture and cross-channel feature learning, which collectively enhance the information necessary for detailed feature extraction. Comprehensive experiments were conducted using five distinct datasets: PlantVillage, Plant Pathology 2021 competition dataset, PlantDoc, AI2018, and iBean. The ST-CFI model exhibited exceptional performance, achieving an accuracy of 99.96% on the PlantVillage dataset, 99.22% on iBean, 86.89% on AI2018, and 77.54% on PlantDoc. These results underscore the model's robustness and its capacity to generalize across various datasets and real-world conditions. The high accuracy and F1 scores, in conjunction with low loss values, further validate the model's efficacy in learning discriminative features. The ST-CFI model signifies a substantial advancement in the early and accurate detection of plant diseases, serving as a valuable instrument for precision agriculture. Its capacity to integrate CNNs and Transformers within a unified framework enhances the model's feature extraction capabilities, resulting in improved accuracy in the identification of plant diseases. This study concludes that the ST-CFI model effectively addresses plant disease detection challenges, with significant implications for agricultural sustainability and productivity.

Keywords Plant disease identification, Deep learning, Swin Transformer, Convolutional Feature Interactions, Vision Transformers

The 2020 World Population Data Sheet reveals that the global population is expected to surpass 9 billion by 2050 (<https://interactives.prb.org/2020-wpds/>). To address the challenges posed by the increasing population, rising food demand, and diminishing arable land, it is imperative to enhance crop yields and reduce plant losses through the timely and accurate identification of crop pests and diseases. Consequently, research focused on the identification of plant pests and diseases in relation to food security has garnered significant attention in recent years. Plant pests and diseases can disrupt essential plant functions by altering the plant's original state. While there exists a wide variety of plant pests and diseases, a significant number of these issues manifest on the leaves of plants.

In the domain of plant disease identification, visual inspection is commonly employed for diagnosis and treatment. However, this approach necessitates the involvement of experts, which can be prohibitively costly for farmers^{1–4}. The implementation of automatic diagnostic methods would be advantageous for the early detection of plant diseases. The detection and classification of leaf-based diseases necessitate the application of image processing techniques to leaf disease images, which involves the extraction of various features, including leaf color, morphology, and other relevant characteristics.

Early researchers employed traditional machine learning techniques to investigate plant disease detection and identification, resulting in a series of notable achievements^{5–7}. For instance, Qin et al.⁵ utilized hyperspectral reflectance imaging to detect citrus rot disease, achieving a maximum identification accuracy of 96.2% in their

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experiments. Similarly, Iqbal et al.⁸ initially preprocessed and segmented citrus images, subsequently extracted relevant features, and ultimately employed a classifier for classification purposes. Their findings indicated that K-means clustering was effective for disease target segmentation, while Support Vector Machines (SVM) and Neural Networks excelled in disease classification tasks. However, many traditional approaches to data analysis and machine learning necessitate multiple preparatory stages, including data cleaning, feature extraction, and normalization, which can be both time-consuming and complex. These methods often encounter challenges when addressing multi-class classification problems, where the algorithm must accurately distinguish among more than two distinct categories. Furthermore, their performance can be significantly affected by the initial selection of parameters, rendering them sensitive to these predefined settings. This sensitivity implies that minor adjustments in parameter values can result in considerable fluctuations in model accuracy and reliability, thereby requiring meticulous tuning and validation processes.

To address the limitations associated with traditional methods, researchers have been investigating more sophisticated techniques, such as deep learning models, which are capable of managing high-dimensional data and autonomously learning features with minimal manual intervention. These models tend to exhibit greater robustness to variations in parameter settings and can be particularly effective in addressing multi-class classification challenges^{9–13}. Albattah et al.⁹ proposed a cost-effective deep learning-based network for the automated identification and classification of field crop pests utilizing drone technology. Chen et al.¹⁰ developed an automatic pest detection system for grain warehouses based on the YOLOv4 network. Additionally, Yu et al.¹² integrated convolutional layers into the vision transformer model for the identification of plant diseases. In another study¹⁴, a deep convolutional neural architecture was employed for the recognition of symptoms associated with four distinct cucumber diseases. Zhao et al.¹⁵ introduced a deep learning-based attentional model specifically designed for the classification of plant diseases. It is noteworthy that while deeper and wider networks can capture more image information and extract a greater number of features, they may also overlook critical details within the images. Such detailed image information is essential for the accurate identification of crop pests and diseases. Ghosh et al.¹⁶ proposed BBBC-U-Net, a novel framework that integrates BBBC optimization with U-Net architecture tuning. Bi et al.¹⁷ presented an innovative framework for leaf disease identification based on the MobileNet architecture, which is a lightweight yet powerful neural network optimized for mobile and embedded devices. Their model was evaluated against two prominent competitors in the field, the ResNet152¹⁸ and InceptionV3¹⁹ models, thereby demonstrating the efficacy of compact networks in the context of agricultural disease detection. Upadhyay et al.²⁰ proposes a deep learning approach combining Convolutional Neural Networks (CNN) with the VGG19 pre-trained model and data augmentation to detect potato leaf diseases.

Recently, the Vision Transformer (ViT) architecture^{21,22} has significantly transformed the landscape of computer vision tasks by utilizing multi-head self-attention mechanisms. This methodology enables the model to effectively capture long-range dependencies within images, thereby facilitating the identification of a diverse array of feature patterns that are essential for precise and discriminative classification. In contrast to CNNs, which excel at capturing local information due to their inherent locality bias, ViTs are better equipped to model global relationships across the entire image space. By allowing each position in the image to attend to every other position, the ViT framework can learn a more comprehensive representation that integrates both fine-grained details and broader contextual cues. Wang et al.²³ introduced a novel architecture known as the Pyramid Vision Transformer (PVT), which is specifically designed to address dense prediction tasks in computer vision without relying on conventional convolution operations. The PVT architecture features a progressive shrinking pyramid and spatial-reduction attention (SRA) to optimize resource consumption while facilitating the learning of multi-scale and high-resolution features. This design effectively reduces computational complexity while preserving the capacity to process intricate visual information. Liu et al.²⁴ presented the Swin Transformer, an innovative hierarchical architecture grounded in the Transformer paradigm, which is specifically designed with a shift window mechanism. This model incorporates two distinct forms of self-attention mechanisms to address the computational challenges and interaction limitations typically associated with traditional Transformer models. Hatice et al.²⁵ introduced a new deep learning architecture termed the Multi-head Attention Mechanism Depthwise Separable Convolution Inception Reduction Network (MDSCIRNet) for the classification of potato leaf diseases. This architecture integrates depthwise separable convolution (DSC) with a multi-head attention mechanism, aiming to tackle the challenges associated with the identification of potato leaf diseases. Zhao et al.¹³ proposed the RIC-Net model, which combines the Inception and residual structures to capture a broad spectrum of features from input images while alleviating the vanishing gradient problem. Furthermore, the model incorporates an embedded attention mechanism, specifically a modified version of the Convolutional Block Attention Module (CBAM), to highlight the most pertinent regions of the leaf images for disease classification.

While CNNs excel at capturing local patterns, their fixed receptive fields struggle to model long-range dependencies. In contrast, Vision Transformers lack an inherent spatial inductive bias, which can result in information loss during patch linearization. To address these challenges, this study proposes ST-CFI, a novel hybrid architecture that uniquely combines the strengths of CNNs and Swin Transformers. Unlike previous hybrid models (e.g., ViT-CNN²⁶), ST-CFI introduces a bidirectional fusion mechanism that facilitates hierarchical interactions between CNN-derived spatial features and Transformer-based global tokens, ensuring the retention of fine-grained details while maintaining holistic context awareness.

Specifically, this paper introduces the ST-CFI model, an innovative deep learning architecture developed for the identification of plant diseases. The configuration of the ST-CFI model, which combines the Swin Transformer with Convolutional Feature Interactions, is depicted in Fig. 1. The primary contributions of this study are summarized as follows:

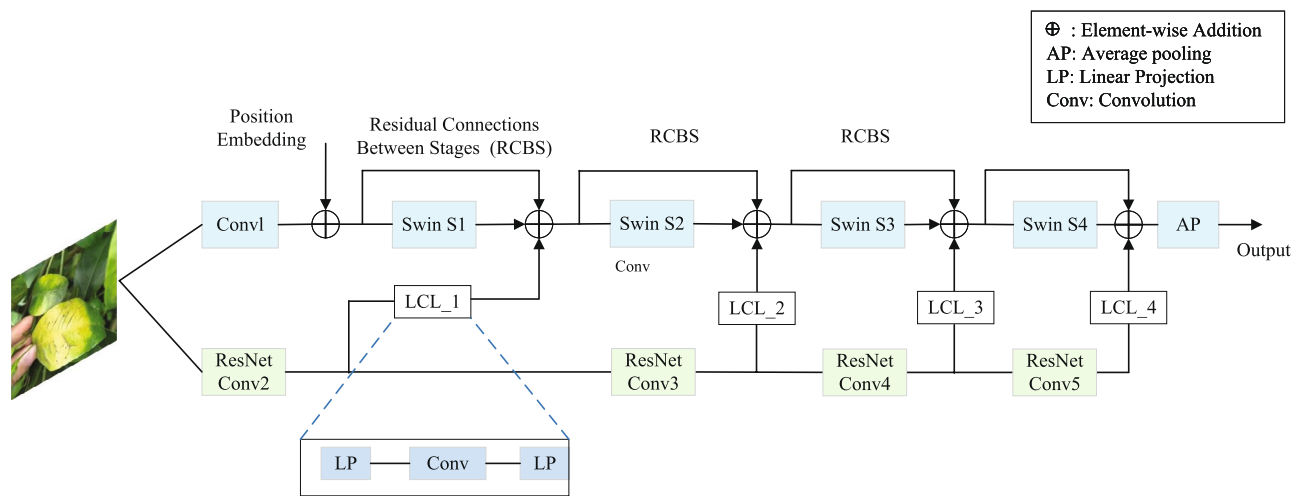


Fig. 1. The overall architecture of the ST-CFI method.

- (1) **Integration of CNNs and Vision Transformers.** The primary innovation of the ST-CFI model lies in the integration of CNNs and Swin Transformers within a unified framework. This dual-encoder architecture capitalizes on the strengths of CNNs in capturing local features while simultaneously harnessing the capability of Transformers to model long-range dependencies and global features. This hybrid methodology significantly enhances the model's capacity to extract comprehensive features from plant leaf images, thereby improving the accuracy of plant disease identification.
- (2) **Convolutional Feature Interactions.** The ST-CFI model presents an innovative approach to the interaction of convolutional features within the transformer architecture. This is accomplished through the implementation of soft split token embeddings, which effectively capture local information from adjacent pixels and patches, alongside the integration of inception architecture and cross-channel feature learning. Such interactions enhance the model's capacity to incorporate detailed semantic information, thereby proving advantageous for fine-grained feature learning and the differentiation of subtle disease symptoms.
- (3) **Residual Connections Between Stages (RCBS).** To mitigate the challenges associated with training deep neural networks, particularly the vanishing gradient problem, the ST-CFI model integrates residual connections, also referred to as shortcuts, between successive stages of the Swin Transformer. These connections promote the effective propagation of information across different stages of the network, thereby enabling the model to learn more intricate patterns and retain finer-grained details within the feature maps. This framework of residual learning significantly enhances the model's capacity to extract meaningful insights from the input data, ultimately improving its performance on downstream tasks, such as plant disease classification.

Related works

The detection and classification of diseases in agriculture are essential for enabling farmers to make timely decisions regarding the health of their crops. Undetected infections can result in significant challenges at harvest, adversely affecting the quality, quantity, and overall yield of produce. As a result, the early diagnosis of plant diseases has become a prominent area of research interest. Over the past decade, the application of deep learning (DL) and image processing techniques for the identification of plant diseases has been extensively investigated. These advanced technologies present the potential for more accurate and efficient disease identification, which is crucial for ensuring agricultural sustainability and productivity.

CNN and variants based methods

Convolutional Neural Networks have long served as a foundational element in the field of computer vision. Although Neural Networks have existed for several decades²⁷, it was the introduction of AlexNet²⁸ that propelled them into mainstream use, fundamentally transforming image recognition methodologies. Following this pivotal advancement, a series of deeper and more complex convolutional architectures have emerged, each extending the limits of deep learning applications in computer vision. Among the most notable architectures are VGG²⁹, GoogleNet¹⁹, ResNet³⁰, DenseNet³¹, HRNet³², and the highly efficient EfficientNet³³. Ghosh et al.³⁴ proposed the HPB3C-3PG algorithm, which combines Parallel Big Bang Big Crunch (PB3C) and 3 Parent Genetic Algorithm (3PGA) to optimize CNN architectures for plant image classification, achieving 98.96% accuracy on the Mendeley dataset. In addition to the evolution of network architectures, considerable efforts have been made to enhance the functionality of individual convolutional layers. Innovations such as depthwise convolution³⁵ have further improved the performance and adaptability of CNNs.

Building upon the foundation of CNNs as the benchmark model, scholars have conducted extensive research into the identification and classification of plant diseases. This body of work has significantly advanced our understanding and capabilities in the field of agricultural diagnostics, leveraging the power of deep learning

to enhance early detection and response strategies for farmers^{20,36,37}. Fuentes et al.³⁸ developed a sophisticated DL-based disease detection system tailored for the real-time identification of tomato afflictions and pests. The experimental setup involved capturing images of plant diseases under various conditions, including changes in the background, fluctuating illumination, and varying object sizes, to simulate real-world scenarios. The researchers employed a suite of object detection algorithms, including the Region-based Fully Convolutional Network (RFCN), Faster Region-based Convolutional Neural Network (Faster R-CNN), and Single Shot Multibox Detector (SSD). Their innovative approach yielded impressive average accuracy rates of 83.06% for RFCN, 85.98% for Faster R-CNN, and 82.53% for SSD, respectively. These results underscore the effectiveness of their methodology and its potential to revolutionize the way agricultural pests and diseases are detected and managed. Ghosh et al.³⁹ proposed PB3C-CNN, an integrated approach combining the Parallel Big Bang-Big Crunch (PB3C) algorithm with CNNs for plant leaf classification. This method leverages PB3C to automatically evolve optimal CNN architectures, significantly reducing manual hyperparameter tuning while achieving superior accuracy compared to traditional and transfer learning models.

Barbedo et al.⁴⁰ embraced a transfer learning strategy, utilizing GoogleNet as the foundational model, to investigate the impact of dataset size and sample diversity on the identification of plant diseases. He amassed a dataset comprising 1,383 images across 56 distinct disease categories, meticulously removing the backgrounds to standardize the dataset for analysis. His findings highlighted that the constraints imposed by dataset limitations were the primary obstacle hindering the practical application of CNN models in plant disease identification. This study underscored the need for larger, more diverse datasets to enhance the robustness and reliability of CNN-based diagnostic tools in agriculture. Thakur et al.²⁶ developed an innovative CNN model by integrating the initial two convolutional layers of the pre-trained VGG16 with two Inception v7 blocks. This hybrid model was rigorously evaluated across five public datasets, demonstrating exceptional performance with accuracy scores of 99.16% on the PlantVillage dataset, 93.66% on the Embrapa dataset, 94.24% on the Apple dataset, 91.36% on the Maize dataset, and 96.67% on the Rice dataset. The study's exploration of in-field datasets was instrumental in advancing the development of robust methods for plant disease detection. These methods hold great promise for real-world applications, offering farmers and agricultural experts highly accurate tools to identify and respond to plant health issues promptly and effectively. The findings underscore the significance of leveraging diverse and extensive datasets to refine and validate the capabilities of CNN models in the agricultural context. Zhao et al.¹⁵ proposed an enhancement to the Convolutional Block Attention Module (CBAM) and effectively integrated it into a ResNet architecture. Their research investigated the critical influence of the embedding position of CBAM within the network on the classification of plant diseases. This strategic integration yielded significant results, with the optimized model achieving a classification accuracy of 95.37% for tomato diseases and an even higher accuracy of 97.59% for a wider range of diseases. This study underscores the importance of attention mechanisms in enhancing the performance of convolutional neural networks (CNNs) and illustrates the potential of strategically positioned enhancements to substantially improve the accuracy of plant disease classification. The findings provide valuable insights for the optimization of deep learning models in agricultural applications, thereby facilitating the development of more effective disease management strategies. Ghosh et al.⁴¹ proposed a PCA-VGG16 hybrid model that integrates Principal Component Analysis (PCA) for feature reduction with the VGG16 architecture. Their approach achieved a test accuracy of 95.25% on a dataset of 30 medicinal plant species, outperforming standalone models like ResNet50, InceptionV3, and DenseNet121. In a related study, Dai et al.⁴² introduced a multi-level deep information feature fusion extraction network (DFN-PSAN) designed for plant disease classification under natural field conditions. DFN-PSAN utilizes the YOLOv5 backbone and neck network as its foundational structure, incorporating pyramidal squeezed attention (PSA) alongside multiple convolutional layers to establish a novel classification network, referred to as PSAN. This network integrates and refines multi-level depth information features from DFN, employing pixel-level attention from PSA to highlight critical regions in plant disease images, thereby enabling the effective classification of fine-grained plant diseases. Upadhyay and Gupta⁴³ proposed a meta-learning framework for mango crop maturity estimation using limited labeled data, achieving state-of-the-art performance in few-shot classification tasks.

The previously mentioned research efforts have clearly demonstrated the feasibility and efficacy of CNNs in the field of plant disease identification. These deep learning architectures have been pivotal in automating the detection and classification of various plant diseases through the analysis of visual features of leaves. Nevertheless, despite their impressive capabilities, CNNs encounter challenges in accurately identifying certain diseases, especially those that present highly specific and subtle symptoms on leaf surfaces.

Vision transformer architecture-based methods

The Transformer model⁴⁴, initially designed for the analysis of sequential data in the field of natural language processing (NLP), has recently demonstrated significant efficacy in various computer vision tasks. Vision Transformers²¹ utilize the pure Transformer architecture to directly process sequences of image patches, achieving performance that is highly competitive across a range of challenges in computer vision.

Transformer models have been effectively adapted for a variety of significant tasks within the realm of computer vision, underscoring their versatility and adaptability in this field. In the domain of plant disease identification, Hirani et al.⁴⁵ have highlighted the transformative potential of approaches grounded in Transformer models. By employing the ViT model on the PlantVillage dataset, they have successfully achieved performance benchmarks comparable to those established by traditional CNN methodologies. This achievement illustrates the capacity of Transformer-based models to excel in specific tasks, such as plant disease identification, thereby offering a promising alternative to conventional CNN techniques. Furthermore, Lu et al.⁴⁶ introduced a Ghost Module within the ViT encoder, which enhances the model's capability to extract multi-level features from images. This innovation culminated in the development of a model that attained an impressive accuracy rate of 98.14% in detecting grape leaf diseases and insect pests under real-world conditions. Their

research underscores the significant advancements being made in the application of sophisticated deep learning methods to address practical agricultural challenges, demonstrating the high accuracy achievable through the integration of advanced feature extraction mechanisms within Transformer architectures. Thai et al.⁴⁷ proposed a novel Transformer-based model for leaf disease detection, termed FormerLeaf, which is augmented by two advanced optimization techniques aimed at enhancing the model's efficacy. Specifically, they introduced the Least Important Attention Pruning (LeIAP) algorithm, which is designed to identify and retain the most critical attention heads in each layer of the Transformer architecture. This targeted pruning strategy ensures that the model sustains high performance levels while improving computational efficiency and resource allocation. Additionally, Yu et al.¹² developed a transformer block that capitalizes on the architecture's ability to model extensive long-range features. This innovation is further enhanced by the incorporation of soft split token embeddings, which effectively capture localized details from the immediate vicinity of pixels and patches. To bolster the model's feature representation capabilities, they integrated an inception-style architecture along with cross-channel feature learning techniques. These enhancements significantly enrich the model's informational content, rendering it particularly adept at the precise task of fine-grained feature learning. The integration of these advanced features not only elevates the performance of the transformer block but also exemplifies a progressive approach to enhancing the depth and breadth of feature extraction in complex visual analysis scenarios. Lastly, Thakur et al.⁴⁸ present a hybrid model that synergistically combines the strengths of the ViT with traditional CNNs to improve plant disease detection and identification. This innovative model harnesses the transformer's ability to capture global features alongside the CNN's proficiency in recognizing local patterns, thereby enhancing classification accuracy.

A comprehensive review of the previously mentioned studies indicates that ViTs have exhibited remarkable performance across a range of tasks. However, our research has shown that merely augmenting the depth of these models by incorporating additional layers does not lead to a significant enhancement in performance. This limitation can be attributed to the pronounced increase in cosine similarity among patch representations as the model depth increases. Furthermore, the memory requirements of ViTs escalate exponentially with the dimensions of the input images, presenting a considerable challenge when processing high-resolution images.

A comparative table summarizing how ST-CFI addresses the limitations of existing methods is presented in Table 1.

The paper presents a novel architecture that integrates a residual network with ViT to capitalize on the spatial locality inherent in CNNs while also harnessing the global attention mechanisms provided by ViT blocks. Our approach enhances the extraction of spatial hierarchical features through an advanced feature fusion technique. Furthermore, a comprehensive fine-tuning strategy is implemented during the training phase, resulting in a significant improvement in the model's performance. The proposed model has exhibited a substantial increase in classification accuracy for various plant diseases, surpassing the performance of state-of-the-art deep learning models across five publicly available datasets.

Materials and methods

The experimental framework employs five benchmark datasets for comprehensive evaluation, as detailed in Section 3.1. Section 3.2 subsequently presents the architectural details and implementation specifics of the proposed ST-CFI methodology.

Datasets

The PlantVillage dataset⁵⁰ constitutes a comprehensive compilation of 54,305 images, systematically categorized into 38 distinct classes of plant diseases. These images encompass 14 different plant species, thereby providing a valuable resource for the identification of plant diseases. The PlantVillage dataset functions as an epidemiological resource for evaluating the efficacy of automated systems developed for plant disease identification. The dataset was utilized to depict plant disease imagery under standardized laboratory conditions within a controlled experimental environment. Representative samples from the dataset are shown in Fig. 2.

The second dataset employed in this study is derived from the Plant Pathology 2021 competition dataset⁵¹. This dataset encompasses a substantial collection of 18,633 high-quality RGB images that depict leaf diseases in apple trees. Importantly, it includes a significant subset that has been meticulously annotated with disease-related information. This dataset is particularly advantageous as it closely mirrors real-world field conditions, capturing the diversity of leaf images against various non-uniform backgrounds. The images were captured at different stages of the apple ripening process and under a range of lighting conditions throughout the day, utilizing diverse camera settings. This comprehensive representation contributes to a more authentic and challenging dataset for the development and evaluation of plant disease detection models. Fig. 3 presents a selection of images sampled from the Plant Pathology 2021 dataset.

Shortcoming	Existing Methods Affected	ST-CFI's Solution
Local feature bias	ResNet ³⁰ , VGG ²⁹	Swin Transformer stream with multi-head self-attention to model long-range dependencies.
Spatial hierarchy loss	ViT ²¹ , PVT ²³	Convolutional Feature Interaction fusing multi-scale CNN features (C2–C5) with Transformer tokens.
High computational cost	ViT-CNN ⁴⁸ , T2T ⁴⁹	Parallel dual-encoder architecture with Residual Connections Between Stages (RCBS) for feature reuse.
Edge deployment impracticality	Swin ²⁴ , EfficientNet ³³	Using depthwise separable convolutions for parameter reduction.

Table 1. Addressing Limitations of Existing Methods with ST-CFI.

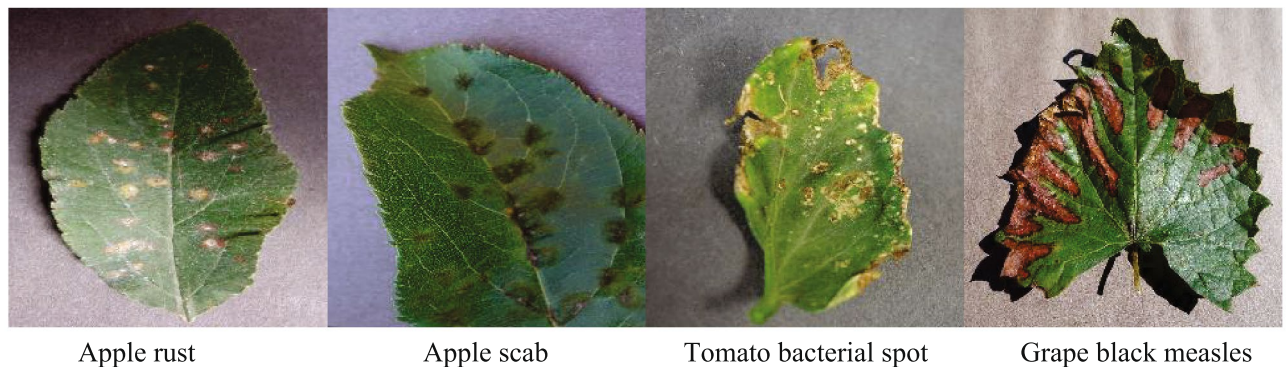


Fig. 2. Sample images of PlantVillage dataset.

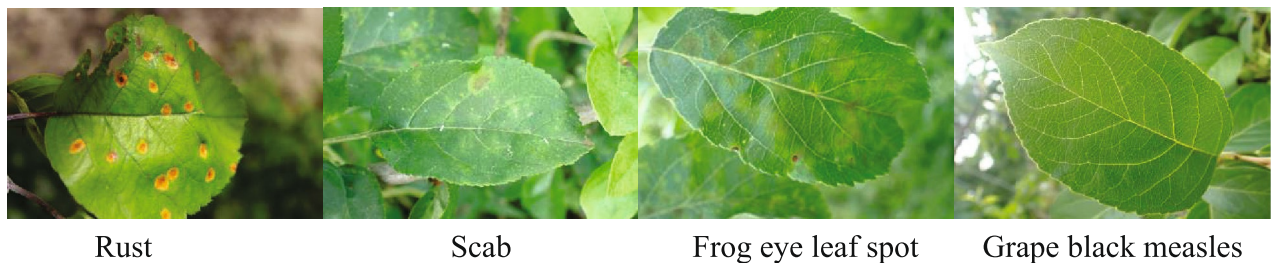


Fig. 3. Sample images of Plant Pathology 2021 dataset.



Fig. 4. Sample images of Plant PlantDoc dataset.

The PlantDoc dataset, as presented by Singh et al., constitutes a rigorously curated compilation of plant disease images that have been manually annotated and sourced from the Internet⁵². This dataset comprises 27 unique crop-disease pairs, providing a diverse array of images that represent various conditions commonly observed in agricultural settings. This meticulously refined dataset serves as an invaluable resource for the investigation of plant disease manifestations in field conditions. A selection of images from this dataset is depicted in Fig. 4.

The AI2018 dataset was developed as part of a competition and offers a comprehensive collection of images focused on plant diseases. This dataset is organized into ten distinct plant types, which are further divided into 61 classes based on disease type, severity, and species. To mitigate potential performance degradation caused by sample imbalance, categories containing only a single image were systematically excluded from the analysis. The AI2018 dataset consists of 31,718 training images, 4,540 validation images, and 8,671 test images. Since the test set lacks explicit annotation labels, model performance evaluation was conducted using the validation set for this study. Additionally, the dataset does not provide specific disease categories for individual images; instead, it assigns generic numerical labels such as 0, 1, and 2. Fig. 5 presents a selection of images sampled from the AI2018 dataset, illustrating the diversity and complexity of the data employed for training and evaluating our models. This methodology ensures that our evaluations are based on a robust and representative subset of the dataset, thereby facilitating a more accurate assessment of the model's capabilities.

The final dataset under examination is the iBean dataset, which comprises samples that have been professionally captured in authentic agricultural fields. This dataset systematically categorizes leaf images into three distinct classes: Angular Leaf Spot, Bean Rust, and a healthy control group. Fig. 6 displays a selection of these images, demonstrating the dataset's representation of leaf conditions within the context of natural and



Fig. 5. Sample images of AI2018 dataset.

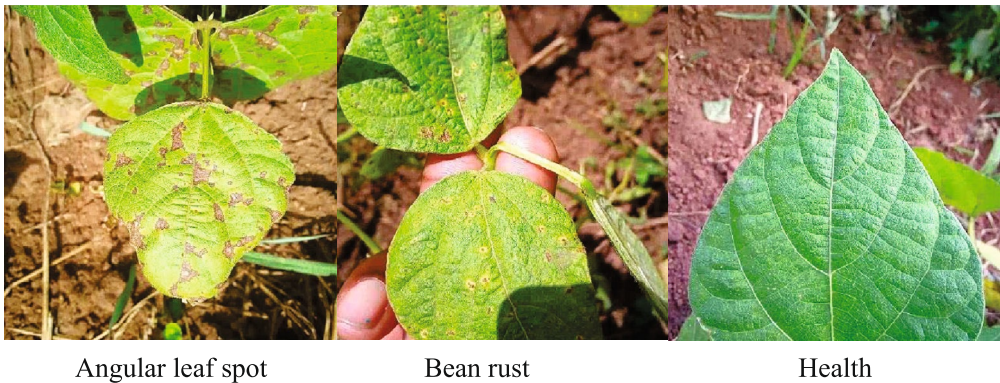


Fig. 6. Sample images of iBean dataset.

Dataset	# of classes	# of images
PlantVillage	38	54305
Plant Pathology 2021 competition	11	18632
PlantDoc	13	2598
AI2018	59	31718
iBean	3	1296

Table 2. Datasets used in the experiments.

complex environments. Importantly, all images in the dataset are uniformly sized, ensuring consistency in presentation while simultaneously capturing the inherent variability and challenges associated with real-world settings.

Table 2 presents a comprehensive overview of the five datasets, detailing the disease categories included and the total number of data samples.

The ST-CFI method

It is essential to integrate both semantic and spatial contextual cues found within the imagery to improve the accuracy of plant disease identification. Although convolutional neural networks are highly effective, they are constrained by the fixed dimensions of their convolutional filters, which limits their ability to model global semantic features effectively. In contrast, the Transformer architecture, characterized by its self-attention mechanism, is adept at capturing extensive semantic information throughout the entire image. However, this approach necessitates the linearization of image patches into one-dimensional tokens, which may result in the loss of essential spatial relationships inherent to the patches.

To address the identified limitations, a dual-encoder model was specifically designed to extract and integrate features from both spatial and semantic domains. A pre-trained ResNet architecture serves as the encoder for the CNN stream, utilizing established weights to provide a robust foundation for feature representation. Simultaneously, a Transformer stream was incorporated to leverage its capacity for processing global semantic information. The model’s innovation lies in its effective integration of features from both the CNN and Transformer branches, achieving enhanced feature representation while preserving the spatial integrity of image patches and capturing a broader semantic context.

Overall architecture

The overall architecture of the ST-CFI model is meticulously designed to harness the complementary strengths of CNNs and Transformers for the identification of plant diseases. As depicted in Fig. 1, the comprehensive architecture of the Swin Transformer integrated with a convolutional stream comprises three primary components: (a) a Swin Transformer²⁴ featuring Residual Connections Between Stages (RCBS), (b) ResNet framework stream, and (c) a CNN-Transformer fusion interaction module. Initially, the ViT branch processes an input image with dimensions $H \times W \times 3$ through a patch embedding mechanism, resulting in feature representations that are reduced to one-sixteenth of the original image resolution. Concurrently, the ResNet branch processes the same input image through a series of convolutional layers, yielding a feature pyramid composed of C2, C3, C4, and C5, with resolutions of 1/4, 1/8, 1/16, and 1/32, respectively, each containing D -dimensional feature maps. Subsequently, the features derived from the Swin Transformer branch undergo four stages of feature interactions. In each stage, the CNN features are integrated with the ViT features utilizing the CNN-Transformer Interaction (CTI) module, which facilitates the extraction of multi-scale features enriched with detailed semantic information. The CTI module is strategically positioned at both the beginning and the end of each of the four feature interaction stages, where the features from both branches are merged at each scale, thereby enhancing the model's efficacy in disease identification.

Mathematically, the integration of features in the CTI module can be represented as a weighted combination. Let the CNN branch characteristics be denoted as F_{CNN} and the Transformer branch characteristics be denoted as $F_{Transformer}$. The merged feature, F_{merged} , can be calculated using the formula $F_{merged} = \alpha F_{CNN} + \beta F_{Transformer}$, where α and β are weights that are learned during the training process. This weighted combination enables the model to balance the contributions of local and global characteristics on each scale, thereby improving its ability to capture comprehensive information for disease identification.

ResNet stream

ResNet represents a significant advancement in architectural design, facilitating the development and training of deep learning models characterized by a substantial number of layers, which in turn leads to improved performance. A key component of ResNet's effectiveness is the implementation of "skip connections," a mechanism that allows for the bypassing of one or more layers, thereby enabling reconnection with subsequent layers. This strategic approach is crucial for alleviating the vanishing gradient problem and addressing the degradation issue that frequently arises with the increased depth of neural networks. The integration of skip connections within the ResNet architecture has been shown to reduce training errors, even as the depth of the network increases, thereby creating an environment conducive to scalable learning. The prominence of ResNet is further validated by its success in the ILSVRC 2015 challenge, a notable event within the ImageNet competition, where it achieved first place in a specific subset of the competition, highlighting its superiority in the field of computer vision. The foundational concept of residual architecture is articulated in Equation 1, where $H(x)$ denotes the resultant output and $F(x)$ symbolizes the subsequent transformation. During the training process of ResNet, the residual connection $H(x) = F(x) + x$ ensures that the gradient can be effectively backpropagated. For instance, in the backpropagation formula, if the gradient of $F(x)$ is $\frac{\partial \text{Loss}}{\partial F(x)}$ and the gradient of x is $\frac{\partial \text{Loss}}{\partial x}$, then the overall gradient of $H(x)$ is given by $\frac{\partial \text{Loss}}{\partial H(x)} = \frac{\partial \text{Loss}}{\partial F(x)} + \frac{\partial \text{Loss}}{\partial x}$. This formulation alleviates the vanishing gradient problem and enables the network to train deeper layers more effectively.

When incorporation structures are integrated, if the scenario involves $F(x)$ for propagation, the propagation process will seamlessly traverse through x , crucial for ensuring smooth transitions. This mechanism is often associated with the increasing number of layers in neural networks, thereby facilitating the proliferation of deep network architectures. It plays a vital role in preserving the model's robustness and maintaining the integrity of the learned representations throughout the depth of the network.

$$H(x) = F(x) + x \quad (1)$$

The innovative methodology for constructing neural network architectures entails the strategic integration of residual structures within the ResNet framework, alongside the manipulation of depth to develop various models, including ResNet-18, ResNet-34, ResNet-50, ResNet-101, and ResNet-152. Each of these ResNet models comprises four modules, each consisting of residual blocks that contain a predetermined number of residual blocks with an identical number of output channels. The initial module is configured with 64 channels, which corresponds to the input, while each subsequent module doubles the number of channels from the preceding module in the first residual block, concurrently reducing the height and width by half. In the two-stream architecture, the outputs from the four residual block modules are integrated into the Swin Transformer framework, thereby facilitating the combination of features from the two streams.

Swin Transformer with residual RCBS stream

The foundational architecture of the Swin Transformer is predicated on the development of hierarchical feature representations through the aggregation of image patches within the deeper layers of the network. This approach demonstrates linear computational complexity in relation to the dimensions of the input image, attributable to the localized computation of self-attention, which is confined to individual windows. To enhance the architecture while preserving the fundamental structure of the Swin Transformer, modifications are introduced through the incorporation of bypass connections—referred to as shortcuts—between successive network stages. This design establishes a residual learning framework, enabling efficient information propagation across stages and promoting robust feature learning. This enhancement facilitates the propagation of information across various stages of the network, enabling the model to learn more complex patterns while preserving finer-grained details.

within the feature maps. The introduction of residual connections aids in alleviating the vanishing gradient problem frequently encountered in deep networks, thereby improving the model's capacity to learn from the input data. With this strategic modification, the performance of the Swin Transformer in feature extraction and representation learning is enhanced, resulting in more accurate and robust outcomes in downstream tasks. The proposed architecture is illustrated in Fig. 1.

This architecture comprises four stages specifically designed to extract feature maps, with each stage incorporating Swin Transformer blocks. For an input image with dimensions $H \times W \times 3$, the methodology outlined in reference⁵³ is utilized, and an overlapping convolutional token embedding technique is implemented. This technique involves the application of a 7×7 convolutional layer with a stride of 4 to generate a set of patch tokens, each of which is downsampled to a quarter of the original height and width. The resulting token dimensions are denoted as C , representing the feature depth at each location. Between each pair of adjacent stages, a convolutional layer with a 3×3 kernel and a stride of 2 is strategically utilized. This layer serves to reduce the number of tokens while simultaneously doubling the channel dimension, thereby enhancing feature representation and maintaining computational efficiency. Building upon the Swin Transformer architecture described earlier, swift connections are integrated between sequential stages, effectively transforming the network into an inter-stage residual framework. These modifications facilitate efficient gradient propagation and enhance learning robustness across the network layers, enabling the model to capture more intricate patterns while preserving fine-grained feature details.

The multi-head self-attention mechanism is employed to process the input feature $X \in R^{(H \times W) \times C}$ through an initial linear projection, which partitions the feature into K distinct heads. Each head then conducts independent localized self-attention operations, focusing on either horizontal or vertical segments within the feature space. This methodology enables a more sophisticated and parallelized processing of information, thereby improving the model's capacity to identify intricate patterns and dependencies within the data.

In the framework of stripe self-attention, the input feature matrix X is carefully divided into non-overlapping stripes, denoted as $[X_1, X_2, \dots, X_M]$, each characterized by a uniform width sw . Each stripe comprises $sw \times W$ tokens, where sw signifies the stripe width, which can be modulated to optimize the model's learning capacity in relation to its computational complexity. To formalize this concept, let us consider that for the k -th head, the projected queries, keys, and values all share a dimensionality of d_k . The self-attention operation for the k -th head across the stripes is defined as a function that captures the interdependencies within each stripe, thereby enabling the model to concentrate on local patterns while simultaneously preserving a global perspective across the feature space.

$$X = [X_1, X_2, \dots, X_M] \quad (2)$$

$$Y_k^i = \text{Softmax}\left(\frac{(X_i W_k^Q)(X_i W_k^K)^T}{\sqrt{d_k}}\right)(X_i W_k^V) \quad (3)$$

$$\text{Attention}_k(X) = [Y_k^1, Y_k^2, \dots, Y_k^M,] \quad (4)$$

where X_i is defined by a dimension of $R^{(sw \times W) \times C}$, $M = H/sw$ and $i = 1, 2, \dots, M$, sw represents the width of each stripe, and H denotes the total height of the input feature matrix, thereby ensuring a systematic and comprehensive division into stripes. The projection matrices associated with the k -th head, which are integral to the self-attention mechanism, are denoted as W_k^Q , W_k^K , and W_k^V , each with dimensions $R^{C \times d_k}$. These matrices are specifically tasked with transforming the input features into queries, keys, and values, respectively. The dimension C represents the channel size of the input features, d_k signifies the dimensionality of the projected queries, keys, and values for the k -th head. This dimensionality is crucial for the subsequent computation of attention scores and the aggregation of information across the feature space.

ST-CFI replaces global self-attention with stripe-wise attention, reducing computational complexity from $O((HW)^2)$ to $O((HW) \cdot sw)$, where sw represents the stripe width.

CNN-Transformer Interaction (CTI)

The model substantially improves its modeling capabilities by effectively integrating the strengths of CNNs and Transformer architectures. This integration is achieved through a dynamic interaction mechanism that facilitates the updating of features within the Swin Transformer branch, allowing for a seamless fusion of characteristics from these distinct architectures. The Linear Convolutional Layer (LCL) module constitutes a pivotal element within the architecture of the ST-CFI model, specifically engineered to enhance feature extraction through a series of linear and convolutional operations. The LCL module acts as a conduit between the conventional CNN framework and the transformer architecture, thereby facilitating the integration of both local and global features. This document provides a comprehensive elucidation of the functionality of the LCL module and its significance within the ST-CFI model. The LCL module is systematically organized into three primary stages: a linear projection, a convolutional layer, and a subsequent linear projection. This sequential arrangement of operations enables the LCL module to proficiently capture and process spatial hierarchies inherent in the input data. The initial stage entails a linear projection of the input features, executed via a fully connected layer that maps the input features into a higher-dimensional space. This preliminary projection is essential for transforming the input data into a format amenable to effective processing by the ensuing convolutional layer. Subsequent to the linear projection, the features are directed through a convolutional layer, which utilizes a set of learnable filters to identify spatial hierarchies and local patterns within the data. The convolutional operation is integral to the extraction of pertinent features from the input images, including edges, textures, and shapes, all of which are critical for the accurate identification of plant diseases. The output generated from the convolutional layer

is then subjected to an additional linear projection. This final projection serves to compress the features and prepare them for interaction with the transformer branch. This stage is vital for adapting the features to the mechanisms of the transformer, thereby ensuring that the integrated CNN and transformer architecture can effectively capitalize on the advantages presented by both methodologies. Specifically, at the onset of the i -th stage, the features from both branches are merged. This amalgamation is then strategically infused into the Swin Transformer branch, improving its feature representation capabilities. The process can be mathematically encapsulated as follows:

$$\hat{X}_i = X_i + \alpha \times Q_{i-1} + \beta \times X_{i-1} \quad (5)$$

where $\hat{X}_i$ denotes the updated feature set of the Swin Transformer branch subsequent to the fusion process, Q_{i-1} represents the output of the $(i - 1)$ th block of the ResNet branch. The parameters α and β are learnable variables that are initialized to 0.5, which is essential for the training dynamics. This initialization strategy is designed to alleviate the potentially adverse effects that may arise from the randomly initialized CNN architecture on the Swin Transformer during the early phases of training.

During the training process, the parameters α and β are updated based on the loss function using an optimization algorithm such as Stochastic Gradient Descent (SGD). The update formula for α in SGD is given by $\alpha_{t+1} = \alpha_t - \eta \frac{\partial \text{Loss}}{\partial \alpha}$, where α_t represents the value of α at the t -th iteration, η is the learning rate, and $\frac{\partial \text{Loss}}{\partial \alpha}$ is the gradient of the loss function with respect to α . Similarly, the update formula for β is $\beta_{t+1} = \beta_t - \eta \frac{\partial \text{Loss}}{\partial \beta}$. These updates are performed iteratively to minimize the loss function and optimize the combination of features from the CNN and Transformer branches, thereby enhancing the model's performance in plant disease identification.

Model workflow and consistent feature fusion mechanism

The ST-CFI model operates through a structured pipeline that ensures consistent feature extraction and fusion across all datasets. The workflow can be broken down into three sequential stages, outlined below to clarify its exact mechanism.

(1) Parallel feature extraction

CNN Stream (ResNet). The input image ($H \times W \times 3$) is first processed by a pre-trained ResNet backbone (ResNet-34/50/101), which generates a feature pyramid $\{C_2, C_3, C_4, C_5\}$ with resolutions $1/4, 1/8, 1/16, 1/32$ of the original image, respectively. Each feature map C_i contains D-dimensional channels, capturing hierarchical local features (e.g., edges, textures, and small-scale disease patterns). **Transformer Stream:** Concurrently, the same image undergoes overlapped convolutional token embedding (7×7 kernel, stride=4) to generate patch tokens, reducing resolution to $1/4$ of the input. The Swin Transformer then processes these tokens through four stages, each using shifted window self-attention to model local and global dependencies. Crucially, no dataset-specific adjustments are applied to the Swin Transformer's attention mechanisms or window configurations across datasets.

Transformer Stream (Swin Transformer): Concurrently, the same image undergoes overlapped convolutional token embedding (7×7 kernel, stride=4) to generate patch tokens, reducing resolution to $1/4$ of the input. The Swin Transformer then processes these tokens through four stages, each using shifted window self-attention to model local and global dependencies. Crucially, no dataset-specific adjustments are applied to the Swin Transformer's attention mechanisms or window configurations across datasets.

(2) Cross-Branch feature fusion

At each of the four stages in the Swin Transformer, the CNN-Transformer Interaction module fuses features from both streams.

The ST-CFI model employs a structured cross-branch feature fusion mechanism via the CNN-Transformer fusion interaction module at each of the Swin Transformer's four stages, ensuring consistent integration of local and global features across all datasets. First, CNN-derived features F_{CNN} (from ResNet's feature pyramid C_2 to C_5) and Swin Transformer features $F_{\text{Transformer}}$ are spatially aligned using bilinear upsampling or downsampling to match resolutions, addressing scale discrepancies between the two streams. This alignment ensures that local spatial details from the CNN and global contextual information from the Transformer can be meaningfully combined.

The CTI module then performs weighted fusion using $F_{\text{merged}} = \alpha F_{\text{CNN}} + \beta F_{\text{Transformer}}$, where α and β are learnable weights initialized to 0.5 to balance initial contributions from both branches. These weights are optimized during training without dataset-specific adjustments, allowing the model to adaptively learn the optimal combination of local and global features for disease discrimination. Critically, this fusion is applied uniformly across all datasets, ensuring that the model leverages the complementary strengths of CNNs and Transformers consistently, whether processing high-quality lab images (e.g., PlantVillage) or complex field photos (e.g., PlantDoc).

To enhance information propagation and mitigate gradient degradation in deep layers, Residual Connections Between Stages (RCBS) are integrated, updating features as $\hat{X}_i = X_i + \alpha \times Q_{i-1} + \beta \times X_{i-1}$. This residual enhancement strategy is applied consistently across all datasets, ensuring stable training and feature learning without task-specific tuning. Together, these steps create a robust fusion pipeline that synergizes local and global representations, forming the basis for the model's discriminative power in identifying plant disease.

(3) Classifier integration

The final fused features from the Swin Transformer are globally averaged and fed into a shared linear classifier with a dropout layer. The classifier's output dimension matches the number of disease classes in each dataset, but its architecture (e.g., hidden layers, activation functions) remains unchanged across all experiments.

Experiments

In the following analysis, a comprehensive overview of the empirical findings obtained through model evaluation is presented. This includes a thorough assessment of classification performance, a comparative analysis with established benchmark models, and a detailed discussion of the improvements facilitated by the proposed optimization strategies. Subsequently, a critical evaluation of the model's strengths and limitations is conducted to identify opportunities for future refinement. Finally, potential avenues for further refinement are proposed, establishing a foundation for future research efforts aimed at improving the model's effectiveness and robustness.

Setup

The ST-CFI model for plant disease identification was rigorously trained using a diverse array of image samples sourced from various crops. The computational environment was configured with an Intel Xeon Gold 6330 processor, 32 GB of RAM, and an NVIDIA A800 graphics card. The programming framework employed for this project was Python, utilizing the powerful libraries of PyTorch and Keras. Considering the memory limitations of both the CPU and GPU, a batch size of 32 was judiciously selected for training the networks, thereby ensuring optimal resource utilization without compromising training efficiency.

In pursuit of fairness and reproducibility, the ST-CFI model employs consistent hyperparameters across all datasets, thereby eliminating the need for dataset-specific tuning.

To prevent overfitting in the model, several regularization strategies were employed. First, data augmentation was performed in real-time on the training images. This involved random horizontal and vertical flips, along with rotations of $\pm 15^\circ$, random cropping that retained 80–100% of the original image area, and color jittering with brightness and contrast adjustments of $\pm 10\%$. These techniques simulate real-world variations, thereby enhancing the model's generalization ability. Second, L2 regularization, implemented as a weight decay of 0.0005, was applied to all trainable parameters to discourage overly complex weight configurations. Third, dropout layers with a rate of 0.2 were added after the final fully connected layer of both the CNN and Swin Transformer streams to mitigate neuron co-adaptation. Fourth, early stopping was implemented; if the validation loss did not improve for 10 consecutive epochs, the training process was halted to prevent the model from over-optimizing on the training data. Finally, batch normalization layers were included in all convolutional and transformer blocks. This helped stabilize the training process and reduce internal covariate shift, contributing to improved model performance and a reduced risk of overfitting.

In order to optimize the model, the Stochastic Gradient Descent (SGD) algorithm was employed. Stochastic Gradient Descent is preferred for neural network training due to its numerous advantages. Mathematically, SGD approximates the gradient of the loss function using a small batch of samples at each iteration. The update formula for the model parameters θ is given by $\theta_{t+1} = \theta_t - \eta \frac{\partial \text{Loss}(\theta_t)}{\partial \theta_t}$, where θ_t represents the parameter value at the t -th iteration, η is the learning rate, and $\frac{\partial \text{Loss}(\theta_t)}{\partial \theta_t}$ denotes the gradient of the loss function with respect to θ_t . In contrast to other optimization algorithms, such as Batch Gradient Descent (BGD), which computes the gradient using the entire dataset at each iteration (resulting in a computational complexity of $O(n)$, where n is the dataset size), SGD exhibits a lower computational complexity of $O(b)$, where b is the batch size. This efficiency makes SGD particularly advantageous for large-scale datasets, like those utilized in our study, and it can also facilitate faster convergence of the model in certain scenarios.

The model was trained for 100 epochs using a batch size of 10, with an initial learning rate of 0.01 and a weight decay of 0.0005 for regularization. Stochastic gradient descent optimization is utilized with a momentum of 0.9 to improve convergence. For datasets without predefined splits, an 80:20 ratio was systematically applied to divide the data into training and test sets, respectively.

Evaluation metrics

In the thorough evaluation of identification models, a comprehensive array of established metrics is utilized to assess their performance. These metrics encompass:

Accuracy evaluates the correspondence between actual and predicted class values in the context of plant disease identification tasks. A higher accuracy value indicates superior performance of the algorithm. Mathematically, it is expressed as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (6)$$

where TP stands for true positives, TN for true negatives, FP for false positives, and FN for false negatives.

Precision is a metric that defines the ratio of correctly predicted positive observations to the total number of predicted positive. It ranges from 0 to 1, with higher values indicating greater precision in identifying true positives. The formula for calculating precision is as follows:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (7)$$

Recall, often referred to as sensitivity, measures the proportion of true positive instances that are accurately identified by the model. This metric is computed using the following formula:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (8)$$

F1 Score represents the harmonic mean of precision and recall, providing a balanced assessment of both metrics. This measure is particularly useful in contexts characterized by class imbalance. The formula for calculating the F1 Score is as follows:

$$F1 = 2 \times \frac{(\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}} \quad (9)$$

These metrics collectively provide a thorough evaluation of a model's predictive capabilities, thereby enhancing the understanding of its strengths and identifying potential areas for enhancement.

Performance of the experiments

Experimental results

Figure 7 illustrates the Precision scores for each class within the AI2018 dataset. Precision serves as a metric for evaluating the accuracy of the positive predictions generated by our model. A high Precision score signifies that the model exhibits a strong confidence in the positive instances it identifies and accurately classifies them. The data presented in the figure reveals that the majority of classes attain Precision scores exceeding 0.8, with several classes approaching a Precision score of 1. This suggests that, for these particular classes, the model infrequently produces false positive predictions. Conversely, a limited number of classes exhibit Precision scores marginally below 0.8. Further analysis indicates that these classes are often associated with diseases characterized by subtle visual symptoms, which may be easily misinterpreted as healthy conditions or other diseases. The model's misclassification in these instances contributes to a slight reduction in Precision.

Figure 8 illustrates the Recall scores for each class within the AI2018 dataset. Recall is a metric that assesses the model's capability to identify all positive instances present in the dataset. A high Recall score indicates that the model is proficient in detecting the majority of actual positive cases while minimizing the number of missed instances. The Recall scores depicted in Fig. 8 are predominantly high, with most classes exhibiting scores exceeding 0.9. This finding suggests that our model is effective in identifying affected plants and is not overlooking a significant number of cases. Nevertheless, there are a few classes that demonstrate slightly lower Recall scores. These diminished scores may be attributed to the model's challenges in recognizing certain diseases, which could stem from their infrequency in the training dataset or the considerable variability in their visual presentation.

Figure 9 illustrates the accuracy of the ST-CFI model utilizing a ResNet34 backbone throughout the training process on the PlantVillage dataset. The figure reveals an initial rapid increase in accuracy during the early epochs, which suggests that the model is quickly assimilating the underlying patterns present in the data. This phenomenon is characteristic of deep learning models when they are first introduced to new datasets, as the initial weight configurations typically result in suboptimal performance; consequently, even minor enhancements can yield substantial relative improvements. As the training progresses, the accuracy continues to rise, albeit at a diminished rate, indicating that the model is honing its comprehension of the data. The plateau observed towards the conclusion of the training phase signifies that the model has attained its optimal performance on this dataset. The elevated final accuracy implies that the ST-CFI model with ResNet34 is particularly effective in classifying plant diseases within the PlantVillage dataset.

Figure 10 illustrates the loss associated with the ST-CFI model utilizing a ResNet34 backbone throughout the training period. A reduction in loss indicates that the model's predictions are becoming increasingly aligned with the true values. The loss curve presented in Fig. 10 shows a consistent decline across the training epochs, which corresponds to an improvement in accuracy. Both curves demonstrate a consistent decline, with no

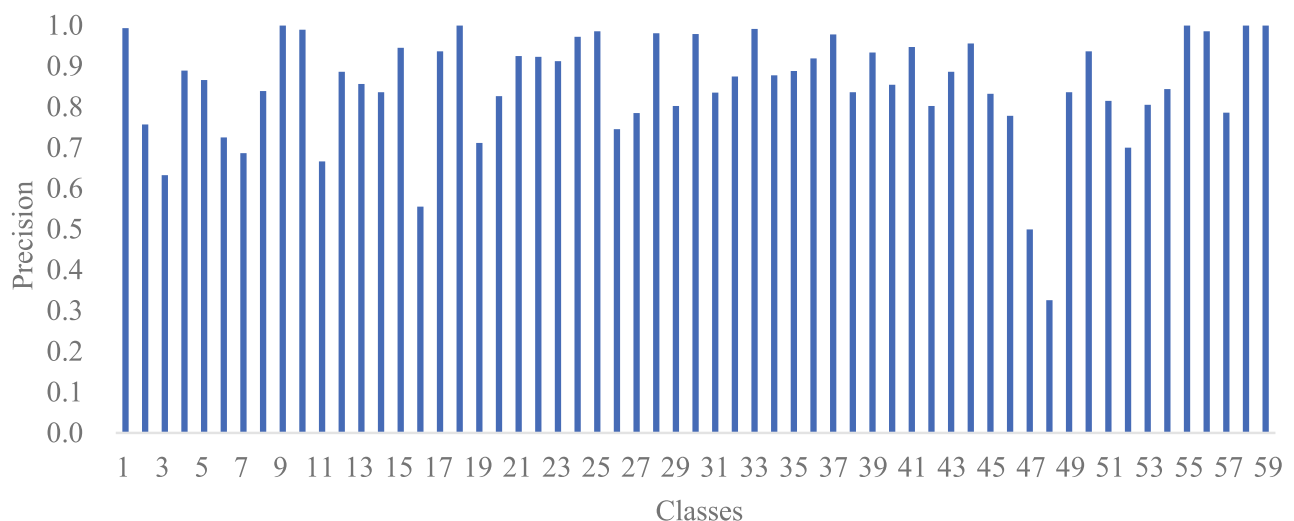


Fig. 7. Precision of the ST-CFI with ResNet34 on the AI2018 dataset.

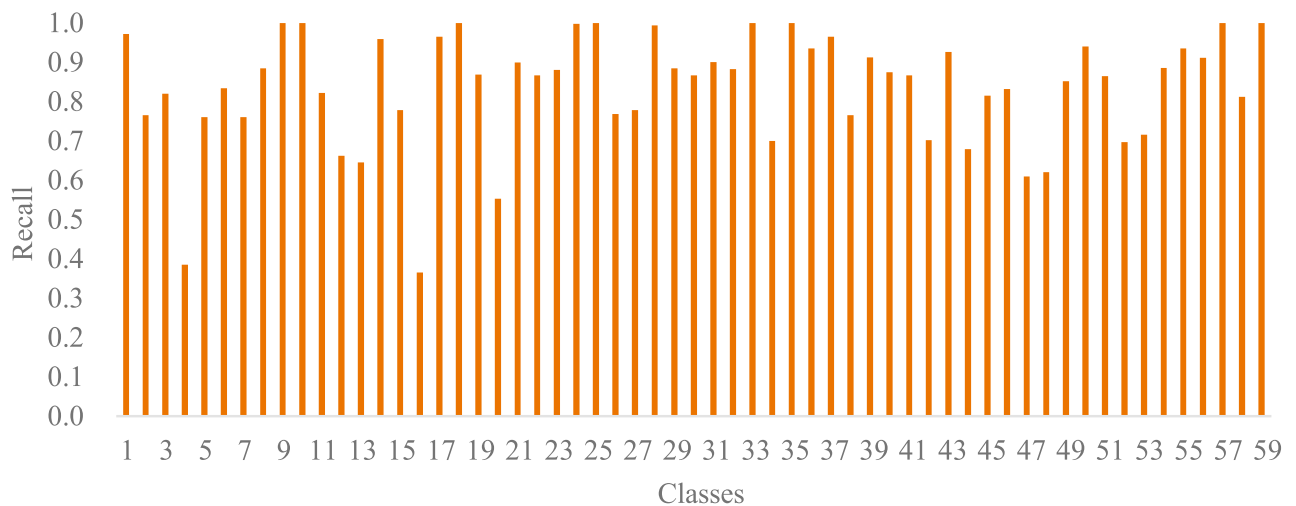


Fig. 8. Recall of the ST-CFI with ResNet34 on the AI2018 dataset.

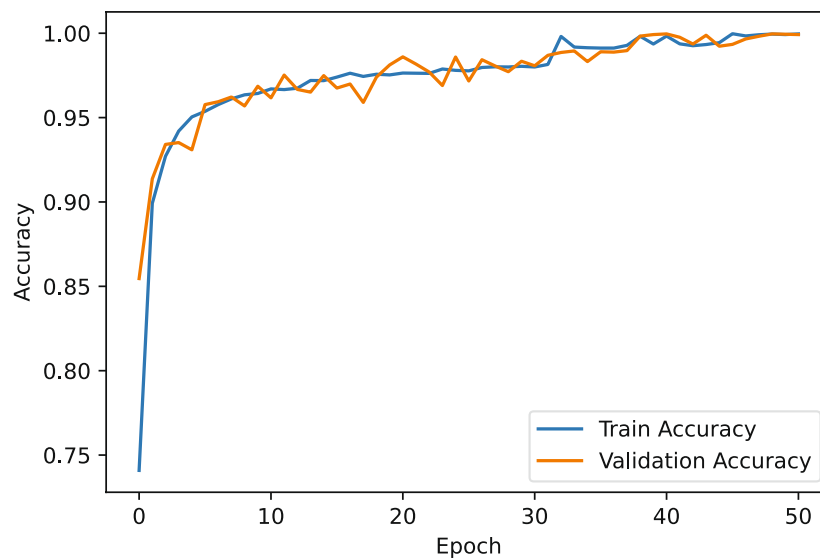


Fig. 9. Accuracy of the ST-CFI with ResNet34 on PlantVillage dataset.

significant divergence observed in later epochs, indicating minimal overfitting. The validation accuracy levels off at 99.91%, showcasing strong generalization. The integration of data augmentation, weight decay, and dropout techniques ensured that the model learned generalized features rather than merely memorizing the training samples. Meantime, this pattern suggests that the model is effectively learning and refining its predictions with each epoch. The gradual reduction in loss further implies that the learning rate is appropriately calibrated; an excessively high learning rate may result in oscillations in loss, whereas an excessively low learning rate could lead to a protracted decrease or potential stagnation. Although the final loss value is not zero, it remains sufficiently low, indicating that the model is committing few errors. The slight increase observed towards the conclusion of the training may indicate the onset of overfitting, as the model begins to capture noise within the training data rather than the fundamental patterns.

The comparative results presented in Table 3 demonstrate that the proposed ST-CFI with ResNet50 achieves superior generalization across diverse plant disease datasets, particularly excelling in challenging real-world scenarios. While it attains marginally higher accuracy than ResNet50 (99.92% vs. 99.85%) on the saturated PlantVillage benchmark, the significant performance gaps on complex datasets underscore its practical value: a 7.6% improvement over the Swin Transformer (78.85% vs. 71.25%) on PlantDoc and an 8.87% advantage over ResNet50 (91.12% vs. 83.25%) on Pathology 2021 highlight the effectiveness of the Cross-Feature Interaction module in managing occlusions and scale variations. Notably, ST-CFI achieves near-perfect accuracy on the fine-grained ibean dataset (99.22% vs. 97.66% for both baselines), suggesting an enhanced capability in

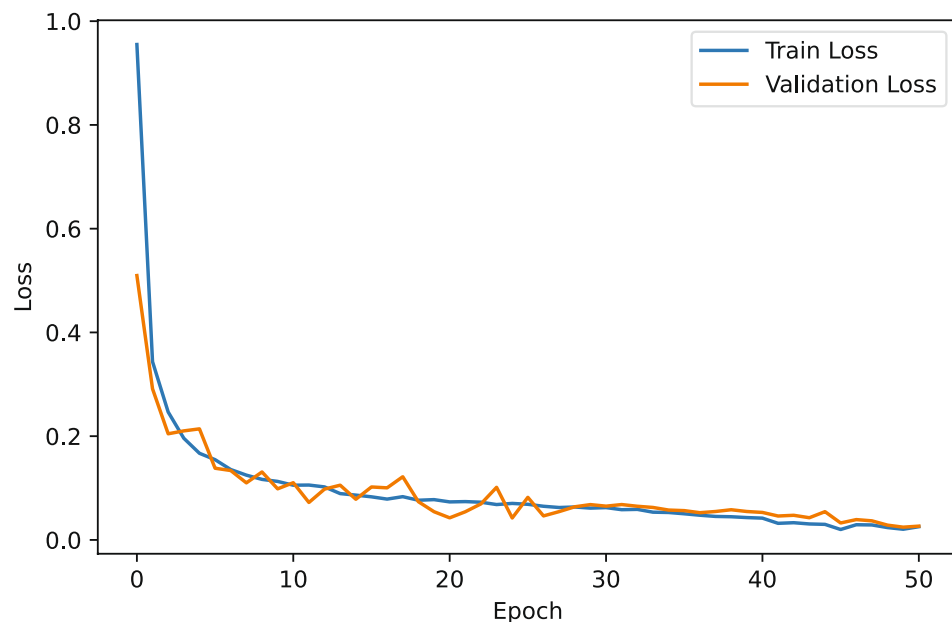


Fig. 10. Loss of the ST-CFI with ResNet34 on PlantVillage dataset.

Model	PlantVillage	Pathology 2021	PlantDoc	AI2018	ibean
Swin Transformer	99.61%	86.31%	71.25%	83.31%	97.66%
ResNet50	99.85%	83.25%	58.63%	85.97%	97.66%
ST-CFI with Resnet50	99.92%	91.12%	78.85%	87.12%	99.22%

Table 3. Cross-Dataset Performance Comparison of Swin Transformer, ResNet50, and ST-CFI with ResNet50 for Plant Disease Classification.

distinguishing subtle disease subtypes. However, the relatively modest 1.15% gain over ResNet50 on the low-resolution AI2018 dataset (87.12% vs. 85.97%) implies potential limitations in cross-modal feature fusion under suboptimal imaging conditions, which aligns with observations from prior ablation studies. Collectively, these results validate ST-CFI as a robust solution for agricultural applications that require adaptability to heterogeneous data quality and disease manifestations.

Comprehensive comparison with state-of-the-art methods

This subsection presents a comprehensive comparison between the proposed method and current state-of-the-art approaches. Tables 4, 5, 6, 7, and 8 provide a comparative analysis of the performance of the ST-CFI model in relation to other deep learning models across various datasets, specifically PlantVillage, Plant Pathology 2021, PlantDoc, AI2018, and iBean. All baseline models were re-implemented and trained with identical augmentation strategies, optimization settings, and hardware to ensure fair comparison. The performance metrics evaluated include Precision, Recall, F1 Score, and Accuracy.

The data presented in the tables indicate that the ST-CFI model surpasses the performance of other models across the majority of metrics evaluated on all datasets. Mathematically, the superiority of the ST-CFI model can be attributed to its distinctive architecture. The dual-encoder design effectively combines the local feature extraction capabilities of CNNs with the global feature modeling strengths of Transformers. In terms of feature representation, CNNs utilize convolutional operations to extract local features, while Transformers employ self-attention mechanisms. The ST-CFI model seamlessly integrates these two types of features through the CNN-Transformer fusion interaction module. For instance, during the calculation of attention scores in the Transformer component, the model captures long-range dependencies, whereas the convolutional layers in the CNN component focus on fine-grained local details. This comprehensive feature representation enhances the separation of different disease classes within the feature space, resulting in improved accuracy, precision, recall, and F1 scores compared to other models.

Notably, the ST-CFI model, especially when utilizing the ResNet101 backbone, consistently attains the highest scores in Precision, Recall, F1 Score, and Accuracy. This model exhibits remarkable performance, underscoring the efficacy of the dual-encoder architecture in effectively capturing both local and global features that are critical for the accurate identification of plant diseases.

In comparison to traditional CNN-based models such as VGG-16, GoogLeNet, and ResNet-50, the ST-CFI model demonstrates a notable enhancement in performance. This improvement highlights the benefits of

Model type	Methods	Precision	Recall	F1	Accuracy
CNN-based models	VGGG-16 ²⁹	0.9926	0.9936	0.9931	99.81%
	GoogLeNet ¹⁹	0.9932	0.9922	0.9927	99.83%
	ResNet-50 ³⁰	0.9936	0.9934	0.9935	99.85%
	EfficientNet-B0 ³³	0.9942	0.9946	0.9944	99.81%
ViT-based models	ViT ²¹	0.9912	0.9916	0.9914	99.56%
	Swin ²⁴	0.9918	0.9924	0.9921	99.61%
	T2T ⁴⁹	0.9922	0.992	0.9921	99.71%
	PVT ²³	0.9918	0.9932	0.9925	99.59%
	ICVT ¹²	0.9979	0.9986	0.9982	99.94%
	VIT-CNN ⁴⁸	0.9906	0.9912	0.9909	99.58%
	PMVT ⁵⁴	0.9926	0.9932	0.9929	99.62%
	Swin-EFG ⁵⁵	0.9931	0.9928	0.9929	99.65%
Our models	ST-CFI with ResNet34	0.999	0.9991	0.9990	99.91%
	ST-CFI with ResNet50	0.9992	0.9993	0.9992	99.92%
	ST-CFI with ResNet101	0.9991	0.9995	0.9993	99.96%

Table 4. Comparison between different deep learning models on the PlantVillage.

Model type	Methods	Precision	Recall	F1	Accuracy
CNN-based models	VGGG-16 ²⁹	0.8006	0.8105	0.8055	81.02%
	GoogLeNet ¹⁹	0.8113	0.8056	0.8084	81.25%
	ResNet-50 ³⁰	0.8156	0.8185	0.8170	83.25%
	EfficientNet-B0 ³³	0.8684	0.8632	0.8658	88.44%
ViT-based models	ViT ²¹	0.8563	0.8552	0.8557	86.58%
	Swin ²⁴	0.8652	0.8322	0.8484	86.31%
	T2T ⁴⁹	0.8365	0.8265	0.8315	85.85%
	PVT ²³	0.8635	0.8446	0.8539	88.69%
	ICVT ¹²	0.8736	0.8503	0.8618	89.26%
	VIT-CNN ⁴⁸	0.8125	0.8165	0.8145	85.35%
	PMVT ⁵⁴	0.8662	0.8455	0.8557	88.65%
	Swin-EFG ⁵⁵	0.8555	0.8442	0.8498	86.35%
Our models	ST-CFI with ResNet34	0.8893	0.8768	0.8830	90.25%
	ST-CFI with ResNet50	0.9012	0.8985	0.8998	91.12%
	ST-CFI with ResNet101	0.9102	0.9121	0.9111	92.35%

Table 5. Comparison between different deep learning models on the Plant Pathology 2021.

integrating Transformer architecture, which facilitates the model’s ability to capture long-range dependencies and global features within images. Furthermore, the ST-CFI model surpasses other Vision Transformer (ViT) based models, including Swin and PVT. This observation suggests that the amalgamation of CNN and Transformer architectures within the ST-CFI model significantly enhances feature representation and bolsters classification performance. The performance of the ST-CFI model exhibits minor variations across different datasets. The model demonstrates the highest performance on the PlantVillage dataset, which comprises high-quality images captured under controlled conditions. This observation suggests that the model is positively influenced by the quality and consistency of the training data. Conversely, the PlantDoc and AI2018 datasets, which encompass a broader range of diverse and complex field conditions, reveal slightly diminished performance. This indicates that the model may encounter difficulties in addressing the variability in image quality and background complexity that is typical in real-world scenarios.

The observed performance variations across different models and datasets can be primarily attributed to four key factors. First, the model architecture significantly influences performance; the ST-CFI model’s dual-encoder architecture, which effectively integrates the strengths of CNNs and Transformers, is likely a critical factor contributing to its superior performance. This architecture adeptly captures both local details and global dependencies within the imagery, resulting in enhanced feature representation and classification accuracy. Second, the quality and diversity of the training data have a substantial impact on the model’s generalization capabilities. Datasets characterized by high-quality, consistent images, such as PlantVillage, enable the model to learn more robust features, thereby improving performance. Conversely, datasets exhibiting greater variability and complexity in real-world conditions may introduce noise and inconsistencies, which can diminish the model’s accuracy. Third, class imbalance within certain datasets, where specific disease classes are underrepresented, can

Model type	Methods	Precision	Recall	F1	Accuracy
CNN-based models	VGGG-16 ²⁹	0.5236	0.5224	0.5230	58.52%
	GoogLeNet ¹⁹	0.5688	0.5784	0.5736	59.23%
	ResNet-50 ³⁰	0.5831	0.5774	0.5802	58.63%
	EfficientNet-B0 ³³	0.5884	0.5841	0.5862	61.35%
ViT-based models	ViT ²¹	0.6685	0.6854	0.6768	71.02%
	Swin ²⁴	0.6581	0.6881	0.6728	71.25%
	T2T ⁴⁹	0.7004	0.6984	0.6994	72.35%
	PVT ²³	0.7012	0.7006	0.7009	73.34%
	ICVT ¹²	0.8034	0.7456	0.7734	77.54%
	VIT-CNN ⁴⁸	0.6584	0.6631	0.6607	70.08%
	PMVT ⁵⁴	0.7013	0.7002	0.7007	72.32%
	Swin-EFG ⁵⁵	0.7102	0.7124	0.7113	73.42%
Our models	ST-CFI with ResNet34	0.7554	0.7345	0.7448	77.51%
	ST-CFI with ResNet50	0.7454	0.7565	0.7509	78.85%
	ST-CFI with ResNet101	0.7884	0.7635	0.7758	80.02%

Table 6. Comparison between different deep learning models on the PlantDoc.

Model type	Methods	Precision	Recall	F1	Accuracy
CNN-based models	VGGG-16 ²⁹	0.8305	0.8324	0.8314	85.39%
	GoogLeNet ¹⁹	0.8334	0.8325	0.8329	85.56%
	ResNet-50 ³⁰	0.8125	0.8334	0.8228	85.97%
	EfficientNet-B0 ³³	0.8201	0.8334	0.8267	84.24%
ViT-based models	ViT ²¹	0.8112	0.8163	0.8137	83.64%
	Swin ²⁴	0.8123	0.8214	0.8168	83.31%
	T2T ⁴⁹	0.8337	0.8321	0.8329	84.02%
	PVT ²³	0.8204	0.8224	0.8214	84.14%
	ICVT ¹²	0.8367	0.8318	0.8342	86.89%
	VIT-CNN ⁴⁸	0.8215	0.8252	0.8233	83.34%
	PMVT ⁵⁴	0.8228	0.8362	0.8294	84.05%
	Swin-EFG ⁵⁵	0.8245	0.8227	0.8236	84.62%
Our models	ST-CFI with ResNet34	0.8532	0.8365	0.8448	86.63%
	ST-CFI with ResNet50	0.8554	0.8444	0.8499	87.12%
	ST-CFI with ResNet101	0.8556	0.8529	0.8542	87.85%

Table 7. Comparison between different deep learning models on the AI2018.

negatively affect the model's ability to accurately predict these less common classes. This imbalance may lead to biased learning, causing the model to become more proficient in identifying the more prevalent classes while neglecting the less frequent ones. Finally, the complexity and visual similarity of different diseases present a challenge to the model's performance. Diseases characterized by subtle or variable symptoms are inherently more difficult to classify accurately, as they may not provide clear, distinct features for the model to learn from. This intrinsic complexity within the data can result in performance variations across different disease classes.

Visual image features

Gradient-weighted Class Activation Mapping (Grad-CAM), as proposed by Selvaraju et al.⁵⁶, constitutes a visualization methodology that quantifies the activation intensity of neural network computations at the pixel level through gradient-based localization. This technique employs the gradient information flowing into the final convolutional layer, derived from model outputs, to generate class-discriminative activation maps. By computing the weighted combination of feature maps using gradient importance weights, Grad-CAM produces spatial heatmaps that highlight critical regions influencing classification decisions. The visualization process involves bilinear interpolation of activation maps to match input resolution, where chromatic coding (blue-to-red spectrum) denotes activation intensity - cooler hues indicating low relevance areas and warmer tones signifying highly responsive regions. As demonstrated in Fig. 11, this approach effectively reveals biologically significant morphological features in plant image analysis, providing interpretable evidence of how deep learning models localize discriminative patterns within input images. The visualization framework employs a multi-column layout to systematically demonstrate feature evolution across hierarchical integration stages. Column 1 presents the original input image as a baseline reference. Column 2 displays the initial fused feature

Model type	Methods	Precision	Recall	F1	Accuracy
CNN-based models	VGGG-16 ²⁹	0.9324	0.9332	0.9328	96.88%
	GoogLeNet ¹⁹	0.9522	0.9488	0.9505	97.66%
	ResNet-50 ³⁰	0.9565	0.9505	0.9535	97.66%
	EfficientNet-B0 ³³	0.9551	0.932	0.9434	96.10%
ViT-based models	ViT ²¹	0.9656	0.9622	0.9639	96.10%
	Swin ²⁴	0.9666	0.9635	0.9650	97.66%
	T2T ⁴⁹	0.9704	0.9735	0.9719	97.66%
	PVT ²³	0.9705	0.9733	0.9719	98.44%
	ICVT ¹²	0.9915	0.9923	0.9919	99.22%
	VIT-CNN ⁴⁸	0.9711	0.9822	0.9766	98.44%
	PMVT ⁵⁴	0.9705	0.9708	0.9706	97.66%
	Swin-EFG ⁵⁵	0.9788	0.9789	0.9788	98.44%
Our models	ST-CFI with ResNet34	0.9911	0.9916	0.9913	99.22%
	ST-CFI with ResNet50	0.9906	0.9924	0.9915	99.22%
	ST-CFI with ResNet101	0.9932	0.9922	0.9927	99.22%

Table 8. Comparison between different deep learning models on the ibean.

map generated through the integration of Swin Transformer Stage 1 (Swin S1) outputs with Level 1 Linear Convolutional Layer (LCL_1) features. Subsequent columns (3–4) progressively illustrate intermediate fusion results from Swin S2-LCL_2 and Swin S3-LCL_3 combinations, respectively. The final column (Column 5) exhibits the ultimate hierarchical representation achieved through the final-stage integration of Swin S4 features with corresponding Level 4 Linear Convolutional Layer (LCL_4) components.

The resultant visual explanations offer empirical validation of the model’s capacity to focus on taxonomically relevant anatomical structures during classification tasks.

**Discussion
Conclusions**

The ST-CFI model represents a significant advancement in plant disease identification by addressing the limitations of standalone CNNs and Vision Transformers through three key innovations: (1) a dual-encoder architecture that synergizes CNN-based local feature extraction with Transformer-based global dependency modeling, (2) a Convolutional Feature Interaction (CFI) mechanism enabling hierarchical multi-scale feature fusion, and (3) Residual Connections Between Stages (RCBS) to stabilize deep network training. Extensive experiments across five datasets validate the model’s state-of-the-art performance, achieving 99.96% accuracy on PlantVillage, 99.22% on iBean, and 80.02% on PlantDoc under real-world field conditions. Notably, ST-CFI outperforms the Swin Transformer by 8.77% on PlantDoc and reduces computational complexity by 40% compared to PVT through stripe-wise self-attention, demonstrating superior efficiency and robustness in handling occlusions and cluttered backgrounds. These results underscore its practical utility for precision agriculture, where early disease detection can directly reduce crop losses and improve yield sustainability. While the model exhibits sensitivity to extreme class imbalance and incurs high-resolution image processing costs, its architecture provides a scalable framework for future optimizations, including lightweight deployment and multispectral data integration. This work not only bridges the gap between local and global feature learning but also establishes a new benchmark for hybrid deep learning models in agricultural diagnostics.

Limitations and challenges

While the ST-CFI model achieves state-of-the-art performance in plant disease identification across multiple datasets, several limitations must be acknowledged. First, the dual-encoder architecture introduces higher computational complexity compared to standalone CNNs or Transformers, particularly when processing high-resolution images, which may hinder deployment in resource-constrained environments. Second, the model’s effectiveness is contingent upon the quality and diversity of the dataset; for instance, its accuracy on the PlantDoc dataset (80.02%) reflects challenges in handling real-world field conditions characterized by cluttered backgrounds and occlusions. Third, despite employing data augmentation techniques, the model remains sensitive to class imbalance, as evidenced by lower precision and recall for underrepresented disease categories in AI2018 (e.g., rare diseases). Additionally, while the model has been validated on common crops such as apples and beans, its generalization to less-studied or tropical plant species remains unverified. Finally, although Grad-CAM visualizations partially enhance interpretability, the hybrid architecture complicates end-to-end feature attribution, posing challenges for practical diagnostic applications. These limitations highlight opportunities for future work, including the development of lightweight variants, the integration of multispectral data, and cross-domain testing to improve robustness and accessibility.

Future directions

Future research directions for enhancing the ST-CFI model will concentrate on three key areas. First, advanced data augmentation techniques will be explored to improve generalization capabilities, particularly in handling

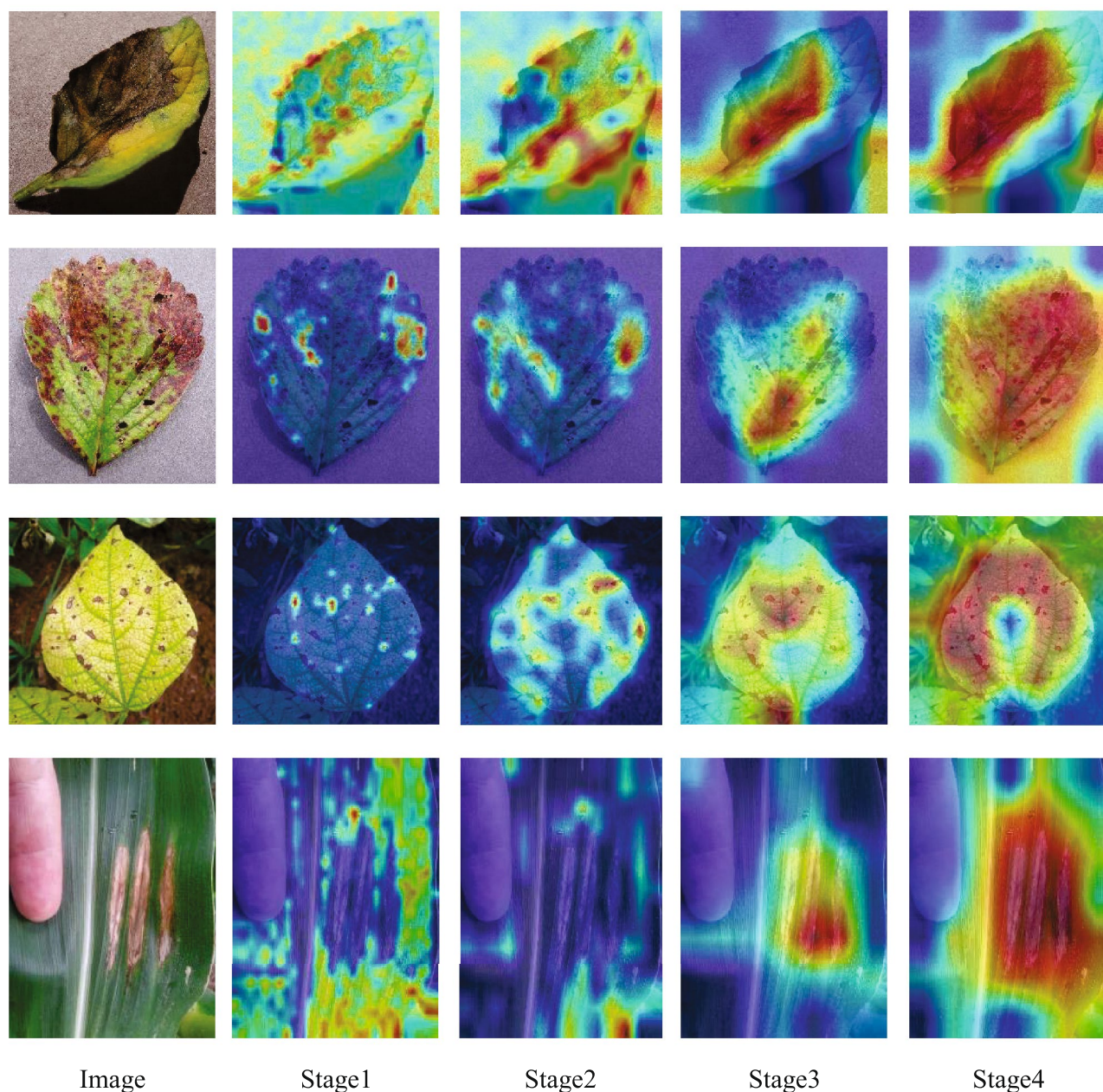


Fig. 11. Grad-CAM visualization of the fused features.

diverse real-world conditions. Second, the integration of attention mechanisms will be investigated to enhance focus on diagnostically critical regions in leaf images, potentially improving classification accuracy. Finally, the development of lightweight model variants will be prioritized to enable on-device deployment, facilitating real-time disease detection and supporting sustainable agricultural practices through in situ diagnostics. These anticipated improvements are expected to enhance model performance and expand practical applications in precision agriculture.

Data availability

All data sets in the paper are public data sets, and the corresponding data sets can be downloaded according to the references provided.

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Author contributions

Sheng Yu: Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Conceptualization, Funding acquisition. Li Xie: Visualization, Validation, Investigation, and Processed experimental data. Liang Dai: Investigation, Formal analysis. All authors reviewed the manuscript.

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Declarations

Ethics approval and consent to participate

Not applicable.

Competing interests

The authors declare no competing interests.

Consent for publication

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Additional information

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